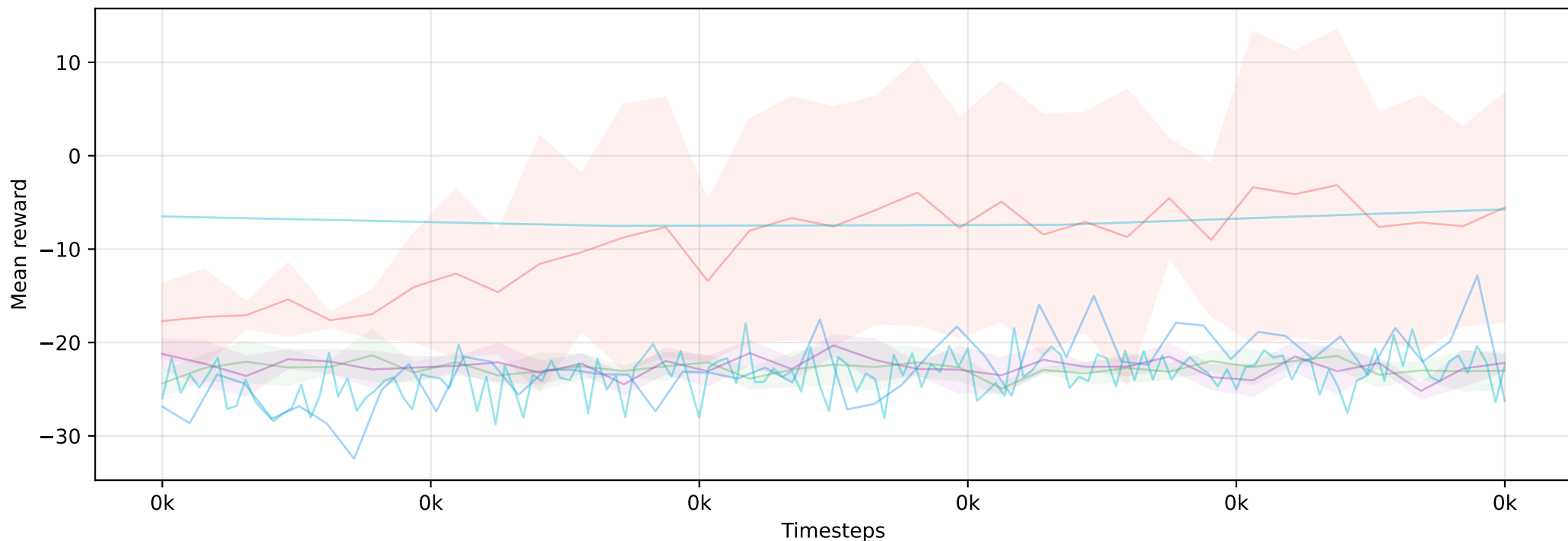
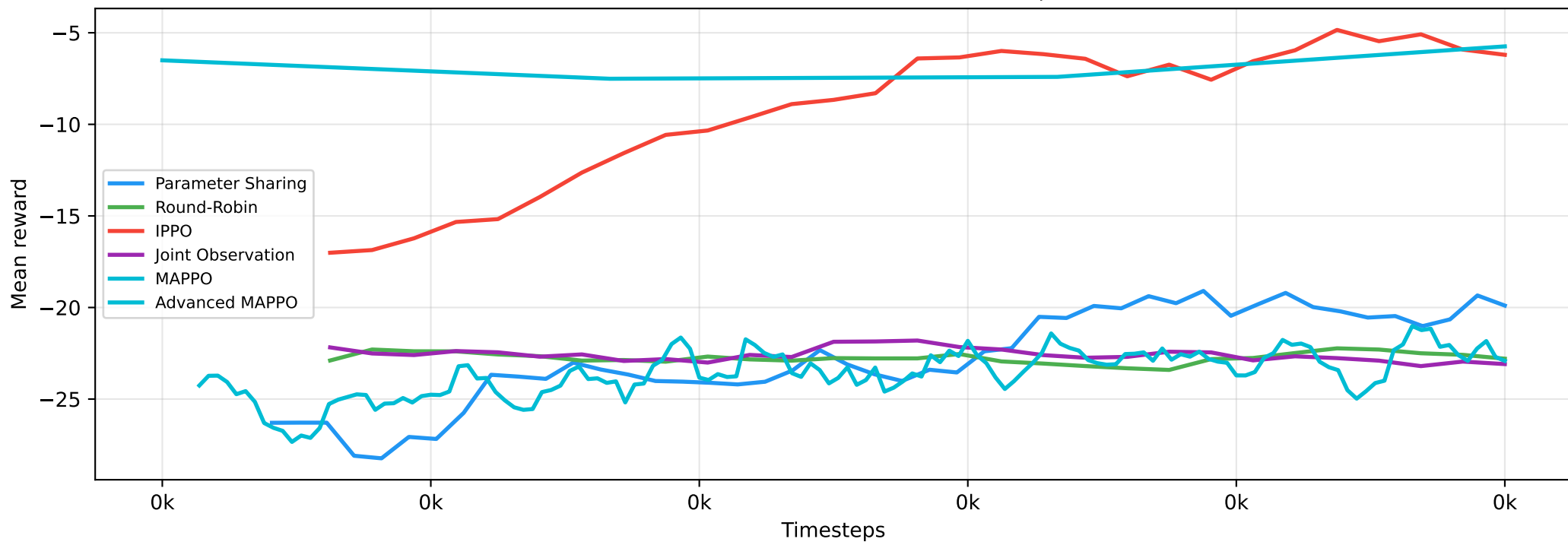


PPO Evaluation — Mean Episode Reward

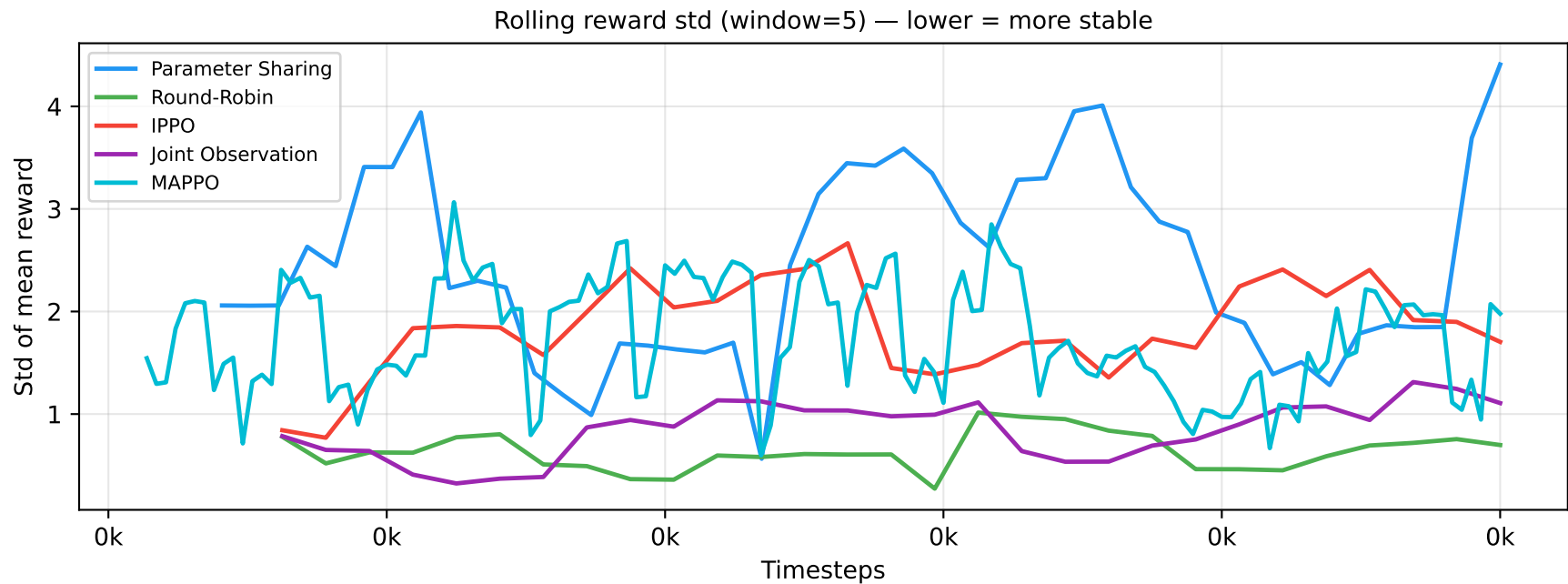
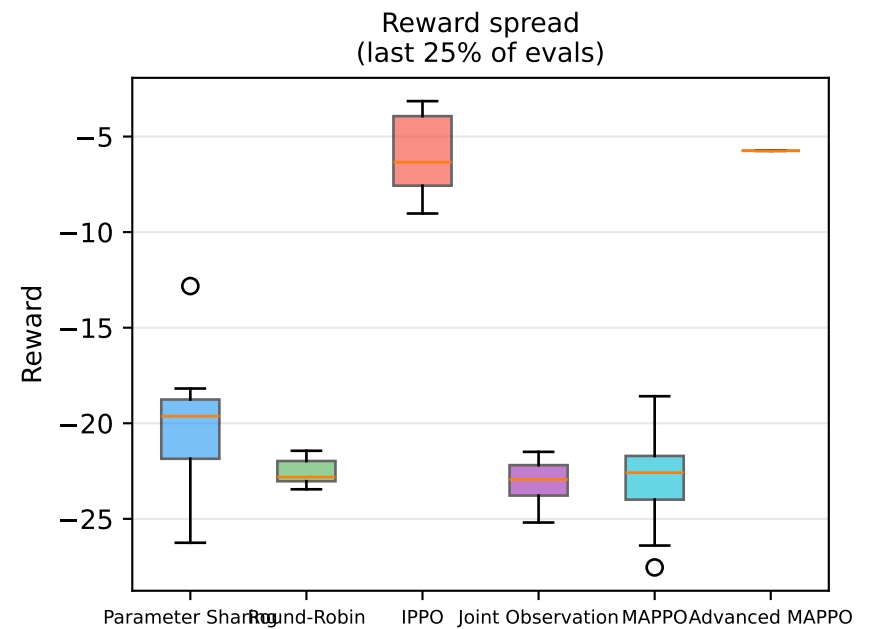
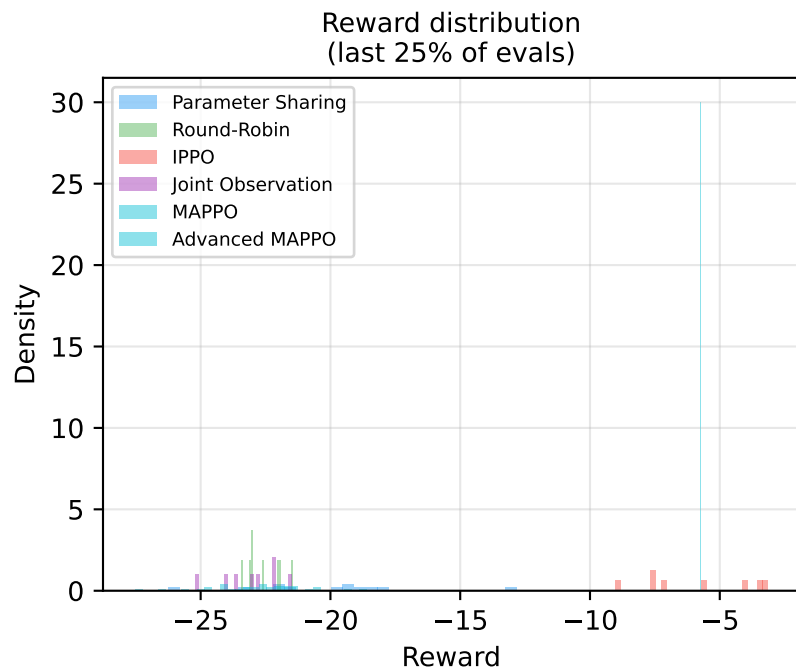
Raw eval mean reward (± 1 std)



Smoothed eval mean reward (window=5 checkpoints)



PPO Evaluation — Distribution & Convergence



PPO Evaluation — Summary Statistics (last 25% of evals)

Run	Total steps	Eval checkpoints	Mean reward	Std	Max	Min
Parameter Sharing	0k	50	-20.183	3.172	-12.822	-26.252
Round-Robin	0k	33	-22.570	0.653	-21.439	-23.454
IPPO	0k	33	-5.942	2.070	-3.148	-9.026
Joint Observation	0k	33	-23.085	1.115	-21.495	-25.192
MAPPO	0k	146	-22.790	1.831	-18.586	-27.542
Advanced MAPPO	0k	4	-5.743	0.000	-5.743	-5.743